Self-Organized Criticality: Sandpile Model on Undirected Graph, Influence of Topology and Dissipation Rules on Avalanches

1. Introduction

In everyday life, person's outburst is rarely caused by a single large event. Rather, the tension builds up from small frustrations over time. When certain treshold is reached, even a minor additional stress can trigger an outburst. That redistributes the tension to others in their social circle. This may in turn cause further reactions, leading to a cascade of reach and size which are difficult to predict. A local event may therefore propagate through a network, depending on individual thresholds and the shape of the network.

This may seem irregular or unpredictable, but many complex systems exhibit stable statistical properties when the number of observation is large. This phenomenon is explained by the law of large numbers [1] which applies, for example, to repeated coin tosses, dice rolls and also many natural phenomenon such as average temperature.

Furthermore, if many independent effects contribute to an outcome, the resulting distribution is often well approximated by a normal distribution according to central limit theorem [1]. A common example is human height which clusters around an *average* with roughly symmetric variation.

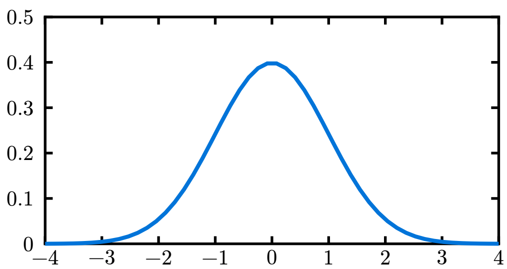

Figure 1: Normal distribution plot.

Contrary to this, there are systems which could not be described by normal distribution [2] . One example are the forest fire sizes [3]. If they were to adhere to normal distribution, there would be a known *average forest fire size*. Even though most forest fires are small, there is nothing limiting the potential size. This is one of the key properties of systems better described by the power law, scale-invariance.

Normal distribution is exponentially bounded, meaning that most of values are distributed within small range around the center (mean) and values far from it are very impropable. The probability density function of normal distribution with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$ is described in Equation (1).

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)} \ . \quad (1)$$

As opposed to this, the power law[1] is *heavy tailed*. Its exponential character described in Equation (2) result in higher probability of extreme values. That corresponds with the scale-invariance: Any (forest fire) size is realistically possible, even though smaller sizes are more likely.

[1]Power law on $(0, \infty)$ cannot be a probability distribution, because the areas near 0 and under the tail are infinite. However, with simple restrictions: $f(x) = ax^{-k}, \quad k > 1, \quad x > x_{\min}$ it is a valid distribution. **TODO: nerozumím poznámce; stačí asymptoticky**

Power law exhibit linear relationship between $\log f(x)$ and $\log x$. In a `log-log plot` it forms a straight line, as can be seen on Figure 2.

$$\begin{aligned} f(x) &= a x^{-k} \\ k &\ldots \text{ constant exponent} \end{aligned} \tag{2}$$

The emergence of the power law distribution is often connected to systems critical state. Consider phase transition of water from liquid to vapor. At the liquid-vapour boundary curve, both liquid and vapour can coexist but are distinctly separate. The boundary terminates at some critical temperature and critical pressure at which the critical point of a system is defined. While the system is in the critical state, the water is not in a single state.

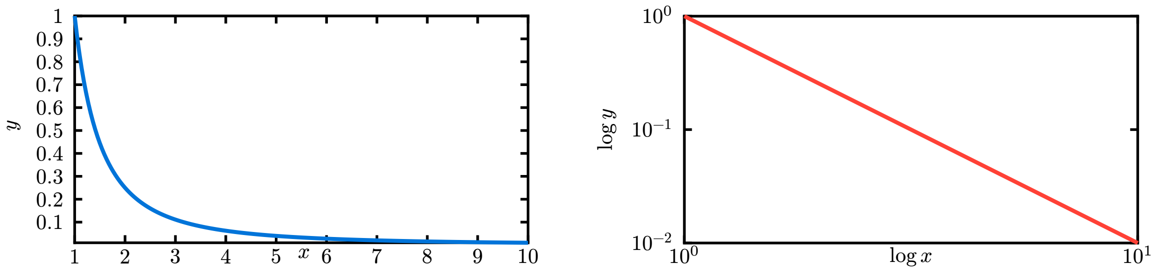

Figure 2: Function $y = x^{-2}$ plotted on linear scale (left) and on log-log scale (right) on interval $\langle 1, 10 \rangle$.

Systems in critical state are most responsive to input and behave unpredictably [4]. This is the case for phase transition and forest fires. However, there is a significant difference. Criticality can be reached in many systems by fine-tuning certain parameters. In the case of phase transition the parameters are pressure and temperature. However, there exist systems which reach criticality independently of parameters.

The phenomenon of systems reaching criticality by themselves is called *self-organized criticality (SOC)*. The concept of SOC was first discovered in 1987 by Bak, Tang and Wiesenfeld and its properties were demonstrated on a sandpile model [5]. The original model was a cellular automaton on a square and cubic lattice.

In this thesis, we investigate the behaviour of the sandpile model on different graph structures. Specifically, on small-world networks which are a class of random networks. Before this generalization we introduce few necessary concepts from graph theory.

2. Graph Theory

An undirected graph is defined as a pair, containing set of vertices and set of edges. The graph has n vertices and m edges.

$$\begin{aligned} G &= (V, E) \\ V &= \{v_1, v_2, \ldots, v_n\} \\ E &\subseteq \{\{u, v\} \mid u, v \in V, u \neq v\}, \quad |E| = m \end{aligned} \tag{3}$$

The neighbourhood of a vertex is a set of vertices connected to it via an edge.

$$\mathcal{N}(v) = \{u \mid \{u, v\} \in E\}, \tag{4}$$

The vertex degree is defined as the size of vertex neighbourhood.

$$d(v) = |\mathcal{N}(v)| \tag{5}$$

Path between two vertices u and v is a sequence of vertices beginning in the first and ending in the second where every two vertices in path must be directly connected.

$$\begin{gathered} P(u,v) = (v_1, v_2, ..., v_k), \quad v_1 = u, \quad v_k = v, \\ \{v_l, v_{l+1}\} \in E \quad \forall l \in \{1, ..., k-1\} \end{gathered} \tag{6}$$

If there exist no such path, let $P_{u,v} = ()$.

Similarily, path length is the number of *steps* (edges) needed to walk the path. If path doesn't exist we define its legth as zero.

$$\operatorname{len}(P_{u,v}) = \begin{cases} |P_{u,v}| - 1 & |P_{u,v}| \geq 2, \\ 0 & \text{else} \ . \end{cases} \tag{7}$$

The shortest path between vertices u and v is the path with minimal length among all paths connecting these vertices. Let $\mathbb{P}(u, v)$ be the set of all paths between u and v:

$$P^*_{u,v} = \min_{P \in \mathbb{P}(u,v)} \operatorname{len}(P) \quad . \tag{8}$$

A graph has an average (shortest) path length $\operatorname{len}(G)$.

$$\operatorname{len}(G) = \frac{1}{n(n-1)} \Sigma_{u \neq v} \operatorname{len}(P^*_{u,v}) \quad . \tag{9}$$

We say that two vertices u, v are connected if at least one path between them exist, i.e.

$$\operatorname{len}(P_{u,v}) > 0. \tag{10}$$

The graph is connected if all its vertices are connected:

$$\forall u, v \in V, u \neq v : \quad \operatorname{len}(P_{u,v}) > 0. \tag{11}$$

The measurement of how much do individual vertices in graph cluster together is called global clustering coefficient and it is based on triplets of nodes. One triplet is defined as three vertices with two or three edges between them. If the number of mutual edges is two, the triplet is called open, if it is three, the triplet is called closed. The global clustering coefficient of graph $C(G)$ is defined as:

$$C(G) = \frac{\# \text{ closed triplets}}{\# \text{ all triplets}} \tag{12}$$

Small world network is a graph with certain characteristic [6]. It has high clustering coefficient and low distances. One example of small world network might be a human relationships network. The low distances might be expressed with respect to number of vertices as:

$$\operatorname{len}(G) \propto \log(|V|) \tag{13}$$

Small world networks tend to contain complete subgraphs (cliques). This is direct result of high clustering coefficient. Additionally, because the average path is small, they often contain *hubs* (vertices with high degree) which serves as connections between other vertices.

One of methods to quantify the *small-worldness* of a network is a small-world measure (ω) defined as follows [7]:

$$\omega(G) = \frac{\operatorname{len}(R)}{\operatorname{len}(G)} - \frac{C(G)}{C(L)} \tag{14}$$

Where G is the analyzed network, R is an equivalent random network to G, and L is an equivalent (ring) lattice network to G. This method aims to quantify the formulating properties of small-world network: high clustering coefficient which is *ideal* in lattice networks, and low average path which is *ideal* in random networks.

3. Sandpile Model

The very original sandpile model is a cellular automaton on a lattice. Every cell can hold up to K grains, the height of a cell is a function of position $z(x, y, ...)$. Fixed boundary conditions are used, on a boundary $z = 0$ cannot be changed. In two dimensions z is updated as follows:

$$\begin{aligned} z(x, y) &\to z(x, y) - 4 \\ z(x \pm 1, y) &\to z(x \pm 1, y) + 1 \\ z(x, y \pm 1) &\to z(x, y \pm 1) + 1 \end{aligned} \tag{15}$$

if z exceeds (or meets) critical value $K = 4$. The system is initialized randomly but with $z \gg K$. That initial state or rather the size of a lattice along with Z (the height of all cells) is called *configuration.*

For simplicity we can imagine a simple chessboard and falling sand grains. Each grain falls on a chess square determined by a random distribution[2]. If the number of grains on any square reaches four, the sandpile topples and the four grains fall on adjacent squares, possibly causing additional squares to topple. In case the square does not have four adjacent squares (i.e. it is on the edge of chessboard) the remaining grains fall off the board and are therefore removed from the model.

We say that cell (square) is stable if it contains less than K grains. The system is stable if all of its cells are stable. If a grain is added to any stable cell, it can destabilize it resulting in an avalanche. Avalanche is the set of topplings which occur before the system is stabilized again.

The sandpile model is a dissipative dynamical system. The dissipation is demonstrating on the boundaries. When a grain leaves the chessboard (or enters a boundary cell) it dissipated from the system. Dissipation is an important property of the model. Without dissipation the model supersaturates, leading to one infinite avalanche.

We can express the square lattice (chessboard) as a graph $G = (V, E)$ with the size given by[3] $\sqrt{n}$. For a boundary condition, the boundary vertices height is fixed to zero $z = 0$. Alternatively, we may imagine only one vertex, called *sink vertex*, with such condition. This sink is connected to all boundary vertices, thus creating the same effect. Please note, that this simplification creates a multigraph, because corner vertices and sink are connected via two edges. For the rest of this thesis we will consider the simplified version with one sink vertex and therefore a multigraph. However, it is trivial to create multiple sink vertices to restore a simple undirected graph.

[2]In the original model, all sand grains were distributed initially. It might be more obvious if we let grains fall one at a time for demonstration purposes.

[3]In a square lattice, the size$^2 = n$, which is the number of vertices in graph.

Figure 3: Comparison of square lattice sink vertices (blue). On the left there is a graph with sinks at the boundary - as in the original model. On the right there is a multigraph, the square lattice with a single sink connected to all vertices at the boundary which is hinted by outgoing edges.

All properties of the cellular automaton system holds on a square lattice. This model was later generalized from square lattice to arbitrary graph [8]. That generalization is important because it allows us to study the model's behaviour for different graph topologies.

3.1. Graph Topologies

Standard choice for sandpile model graph is square lattice with arbitrary size. This however do not realistically represent the model from motivation story in Section 1. That is why we introduce different graph topologies.

Before leaving square lattice entirely, we discuss toppling and dissipation rules on it. The original model presents to us a toppling rule described in Equation (15) and $K = 4$, where K is the number of grains which cause a single cell (vertex) to topple.

Dissipation rules on general graph are more complex and further discussed later in Section 3.2. While the different definitions of K and toppling rules might be more deeply explored, we will only consider K being a function of vertex:

$$K(v) = d(v), \quad v \in V. \tag{16}$$

And the toppling rule is also expressed as a function of vertex. Similarily to original definition as cellular automaton, the height of a vertex is expressed as a function $z(v)$:

$$\begin{gathered} z(v) \to z(v) - K(v) \\ z(n) \to z(n) + 1, \quad n \in \mathcal{N}(v) \end{gathered} \tag{17}$$

This simple rule states, that during the toppling process the original vertex loses as many grains as many neighbours it has and all its neighbours gain one. The benefit of this K and toppling rule is their simplicity.

For example, it might be interesting to set $K = \text{const}$, but that would create problems. If a vertex has less neighbours, it is unclear where the additional grains go during a topple, and if it has more neighbours, does it artifically generate more grains, or randomly distribute fewer than K.

Note that in these reasonings, the neighbourhood of a vertex $\mathcal{N}(v)$ contains sink vertex, and possibly multiple times. The toppling rule will not work if $K(v) = 0$. We acknowledge this constraint but leave its resolution to dissipation rules.

With K, toppling rule and dissipation rules specified, we now continue onto the properties of different graph topologies. Sqaure lattice is a regular graph, meaning the degree of all vertices is the same [9], if we consider sink vertex.

Regularity is not often found in human social networks, meaning graphs where vertices represent individuals and edges their mutual acquaintance. Purely random graph also would not be optimal for although it is not regular, human networks are not random. Fortunately, a small-world network (better described in Section 2) are almost perfect.

The three most known small-world networks are the Erdős–Rényi model [10], Barabási-Albert [11] model and Watts-Strogatz model [6].

TODO: ER graph

TODO: BA graph

TODO: WS graph - we have (why)

3.2. Dissipation Rules

With the introduction of arbitrary graph we must reconsider dissipation rules. The square lattice has explicit boundaries which serve as sinks. However, many graphs do not have an implicit boundary. The existence of sink is fundamental for the model not to supersaturate.

A dissipation rule determines how many times a vertex should be connected to the sink. There is not a single rule which would be generally applied to any graph. We discuss different rules, their properties and analogies. Dissipation rules are not well-known thus their names are created for the purposes of this thesis.

We can present a dissipation rule as a function of vertex returning an integer which denotes how many times a vertex should be connected to the sink.

$$D : v \to \mathbb{N} \quad . \tag{18}$$

For example, the graph representation of the original model uses a *Fill to Four* rule. It states that the degree of any vertex must be four or more. The sink vertex is connected to all vertices as many times it is needed to fulfil that rule. For the square lattice, only vertices on the edge are connected (exactly once except for corner vertices, which are connected twice).

$$D(v) = \begin{cases} 4 - d(v) & d(v) \geq 4, \\ 0 \text{ .} \end{cases} \tag{19}$$

For different graphs though, this rule might be insufficient. It does not ensure dissipation. This rule does not consider the vertex degree if it is higher than four.

To ensure dissipation on any graph, the rule must connect all vertices to the sink at least once. That is because there might exist vertices which are not connected to any other vertex. A problem arises with such vertex if it were not connected: $d(v) = 0 = K$ meaning it should trigger an avalanche but that itself is nonsensical. Dissipation rules must prevent this from happening.

If we consider connected graphs the dissipation condition softens. It is guaranteed that for all pairs of vertices exist at least one path therefore only one sink is sufficient to ensure dissipation. This might be useful for the sandpile model is only interested on connected graphs where they exhibit critical behaviour.

One possible rule which guarantees dissipation on any graph is *All Once*. It ensures just the necessary condition that every vertex must be connected to sink at least once but nothing more. The rule is fair to all vertices as the connection to the sink is not exclusive to vertices of some characteristic[4].

$$D(v) = 1 \quad . \tag{20}$$

[4]Possible characteristics include vertex neighbourhood size, shortest path by which it reaches all other vertices and more.

One rule sufficient for connected graph is *As Many As Neighbours*. This rule connects every vertex to sink proportionally to its neighbourhood size. When considering connected graph this rule ensures dissipation and even provide a connection to the sink to all vertices.

$$D(v) = d(v) \quad . \tag{21}$$

Every rule represents a different approach and not all can be applied to all types of graph topology. **TODO**

TODO: put graph topology & dissipation rules together

4. Observed Behaviour

TODO: how did it work for different types of graphs and dissipation rules? also the graph configurations (eg β for WS)

TODO: metrics, how it changed criticality? origin distribution, avalanche size (max, avg)

TODO: imagery from the app

5. Conclusion

TODO

6. Technical Documentation

TODO

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